[Aim: 100/100 in Maths]

आरंभ

REAL NUMBERS

Lecture #8

LP: Three bells ring at intervals of 6, 12 and 18 minutes. If all three bells rang at 6 AM. When will they ring together again.

6 min

12 min

18 min

$$\text{LCM}(6, 12, 18 \text{ min}) \Rightarrow \boxed{36 \text{ min}}$$

6:00 am

$\downarrow$

$\boxed{6:36 \text{ am}}$

Ans

Irrationality : (Q')

$\sqrt{2} \rightarrow Q'$ ✓

$\sqrt{3} \rightarrow Q'$ ✓

~~$\sqrt{4} \rightarrow \boxed{2 \rightarrow Q}$~~

~~$\sqrt{9} \rightarrow 3 \rightarrow Q$~~

$\sqrt{\text{(not a perfect square)}}$

Irrational no

$\sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{7}, \sqrt{6}, \sqrt{8} \ldots$

Rational no. ($\mathbb{Q}$) $\longrightarrow$ $\frac{p}{q}$ form ($q \neq 0$)

$\longrightarrow$ $\frac{p}{q}$ (p & q doesn't have any common factor)

$\frac{1}{2}$ ✓

$\frac{\cancel{5}^{1}}{\cancel{10}_{2}} \Rightarrow \frac{1}{2}$

$\frac{\cancel{6}^{3}}{\cancel{8}_{4}} \Rightarrow \frac{3}{4}$

$\frac{\cancel{14}^{2}}{\cancel{49}_{7}} \Rightarrow \frac{2}{7}$

Axiom : if p divides a^2, then p divide a also.

~~eg~~ if 2 divides $(4)^2$

then 2 divide 4 also

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if 4 divides $(36)^2$

then 4 divide 36 also.

LP: Prove that is √2 is an irrational number. [by contradiction]

Let $\sqrt{2}$ is rational no.

$\sqrt{2} = \frac{p}{q}$ (p & q doesn't have any common factor)

squaring both sides

$(\sqrt{2})^2 = \frac{p^2}{q^2}$

$2 = \frac{p^2}{q^2}$

$\boxed{q^2 = \frac{p^2}{2}}$ — I

2 divides p^2

then 2 divide p also

(let) $n = \frac{p}{2}$

$\boxed{p = 2n}$

put in I

$q^2 = \frac{(2n)^2}{2}$

$q^2 = \frac{\cancel{4}^{2} n^2}{\cancel{2}}$

$\frac{q^2}{2} = n^2$

2 divides q^2

then 2 divide q also.

(let) $m = \frac{q}{2}$

$\boxed{q = 2m}$

$\frac{p}{q} = \frac{2n}{2m}$

this contradiction is due to wrong assumption that $\sqrt{2}$ is rational.

Hence, $\sqrt{2}$ is irrational.

LP: Prove that is √5 is irrational.

Let $\sqrt{5}$ is rational no.

$\sqrt{5} = \frac{p}{q}$ (p & q doesn't have any common factor)

sq. both sides

$(\sqrt{5})^2 = \frac{p^2}{q^2}$

$5 = \frac{p^2}{q^2}$

$\boxed{q^2 = \frac{p^2}{5}}$ — (I)

5 divides p^2

5 divide p also

Let $n = \frac{p}{5}$

$\boxed{p = 5n}$

put in (I)

$q^2 = \frac{(5n)^2}{5}$

$q^2 = \frac{\cancel{25}\,5 n^2}{\cancel{5}}$

$q^2 = 5n^2$

$\frac{q^2}{5} = n^2$

5 divide q^2

5 divide q also

(let) $m = \frac{q}{5}$

$\boxed{q = 5m}$

$\frac{p}{q} = \frac{5n}{5m}$

here 5 is a common factor.

this contradiction is due to our wrong assumption.

Hence, $\sqrt{5}$ is irr.

HP

Q. Prove that $\sqrt{7}$ is irrational no.

Let $\sqrt{7}$ is rational no.

$\sqrt{7} = \frac{p}{q}$ (p & q doesn't have any common factor)

Sq. both sides

$(\sqrt{7})^2 = \frac{p^2}{q^2}$

$7 = \frac{p^2}{q^2}$

$\boxed{q^2 = \frac{p^2}{7}}$ — (I)

7 divide p^2

then 7 divide p also.

(let) $n = \frac{p}{7}$

$\boxed{p = 7n}$

put in (I)

$q^2 = \frac{(7n)^2}{7}$

$q^2 = \frac{\cancel{49}\,n^2}{\cancel{7}}$

$\frac{q^2}{7} = n^2$

7 divides q^2

7 divide q also

$\frac{q}{7} = m$ (let)

$\boxed{q = 7m}$

$\frac{p}{q} = \frac{7n}{7m}$

7 is a common factor

this contradiction is due to our wrong assumption.

Hence, $\sqrt{7}$ is irrational.

Hence Proved.

#K^3B

R, R	R, I_r	I_r, I_r
$R+R$	$R+I_r$	I_r+I_r
$R-R$	$R-I_r$	I_r-I_r
$R\times R$	$R\times I_r$	$I_r\times I_r$
$\frac{R}{R}$	$\frac{R}{I_r}$	$\frac{I_r}{I_r}$
R	I_r	Cannot say

$(R \neq 0)$

LP: Prove that is $3\sqrt{5}$ is irrational. Given that $\sqrt{5}$ is irrational

Let $3\sqrt{5}$ is rational.

$$3\sqrt{5} = \frac{p}{q}$$

$$\sqrt{5} = \frac{1}{3} \times \frac{p}{q}$$

$\sqrt{5} \rightarrow$ Irr (given)

$\underbrace{\frac{1}{3}}_{R} \times \underbrace{\frac{p}{q}}_{R} \rightarrow R$

but Irr $\neq$ R

this contradiction is due to our wrong an.

Hence $3\sqrt{5}$ is irr. ~~HP~~

Next Toppers

#LP: Prove that $4 - 2\sqrt{5}$ is an irrational number given that $\sqrt{5}$ is irrational. **[CBSE 2026]**

Let $4 - 2\sqrt{5}$ is rational.

$$4 - 2\sqrt{5} = \frac{p}{q}$$

$$-2\sqrt{5} = \underbrace{\frac{p}{q} - 4}_{R}$$

$$\underbrace{\sqrt{5}}_{\text{Irr. (given)}} = \frac{\frac{p}{q} - 4}{-2} \rightarrow R$$

but Irr $\neq$ Rat.

this contra... - ——

- ——

#LP: Prove that $(\sqrt{2}+\sqrt{3})^2$ is an irrational number, given that $\sqrt{6}$ is an irrational number.

Let $(\sqrt{2}+\sqrt{3})^2$ is rational.

$$(\sqrt{2}+\sqrt{3})^2 = \frac{p}{q}$$

$$(\sqrt{2})^2 + (\sqrt{3})^2 + 2\sqrt{2}\times\sqrt{3} = \frac{p}{q}$$

$$2+3+2\sqrt{6} = \frac{p}{q}$$

$$5+2\sqrt{6} = \frac{p}{q}$$

$$2\sqrt{6} = \frac{p}{q} - 5$$

R R

$\hookrightarrow$ R

$$\sqrt{6} = \frac{\frac{p}{q} - 5}{2}$$

R R R

given Irr.

but Irr $\neq$ R

This contra - - —

#LP:Show that $\frac{1}{\sqrt{3}+\sqrt{2}}$ is irrational, given that $\sqrt{6}$ is irrational.

$\Rightarrow \frac{1}{\sqrt{3}+\sqrt{2}} \times \frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}-\sqrt{2}}$

$\Rightarrow \frac{\sqrt{3}-\sqrt{2}}{(\sqrt{3})^2-(\sqrt{2})^2}$

$\Rightarrow \frac{\sqrt{3}-\sqrt{2}}{3-2} \Rightarrow \frac{\sqrt{3}-\sqrt{2}}{1}$

$\Rightarrow \sqrt{3}-\sqrt{2}$

Prove $(\sqrt{3}-\sqrt{2})$ is irr.

Let $\sqrt{3}-\sqrt{2}$ is rational

$\sqrt{3}-\sqrt{2} = \frac{p}{q}$

squaring both sides

$(\sqrt{3}-\sqrt{2})^2 = \frac{p^2}{q^2}$

$\Rightarrow (\sqrt{3})^2+(\sqrt{2})^2-2\sqrt{3}\sqrt{2} = \frac{p^2}{q^2}$

$\Rightarrow 3+2-2\sqrt{6} = \frac{p^2}{q^2}$

$\Rightarrow 5-2\sqrt{6} = \frac{p^2}{q^2} \longrightarrow$ HW

Real No. समाप्ति !!

THANK YOU COODIESS !!

HW → Complete Module.